Week 7, Lesson 7

AWS Data Analytics Services

Chapter 5 – *Amazon Data Analytics Services* (Song, 2023, pp. 105–144)

Outline

1. Understanding the AWS big data pipeline
2. AWS Glue, Athena, and Kinesis family
3. Amazon EMR, Hadoop, and MapReduce
4. Visualization with Amazon QuickSight

Dr. Rohit Kumar Kasera
Lecturer

Understanding the AWS big data pipeline

Why Data Analytics in the Cloud?

Organizations generate massive data every second from:

- Web applications
- Mobile devices
- IoT sensors
- Social media
- Business transactions

What is Big Data ?

Big data refers to extremely large and complex datasets that traditional systems cannot process efficiently.

Why AWS?

Benefits of Cloud Analytics

- Scalability
- Elasticity
- Managed services
- Pay-as-you-go pricing
- Global infrastructure

Business Value

- Faster decisions
- Real-time insights
- Reduced infrastructure cost
- Better customer experience

3Vs of Big Data

- **Volume** → Massive amounts of data
- **Velocity** → Fast data generation
- **Variety** → Different data formats

Examples

- Netflix recommendations
- E-commerce clickstreams
- Financial transactions
- Social media feeds

From Raw Data to Insights

Data Analytics Pipeline Flow

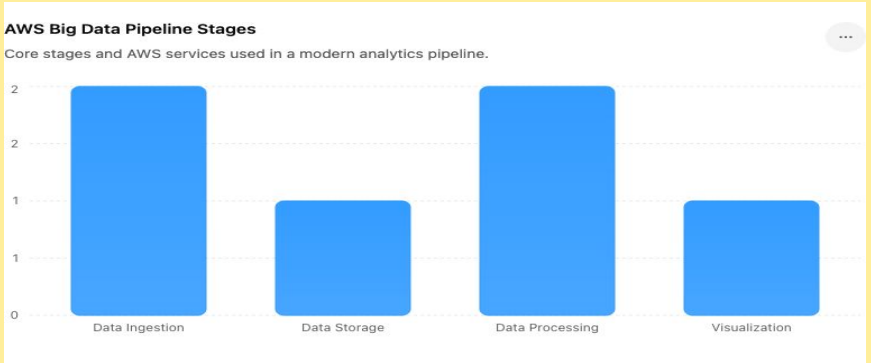
Raw Data → Storage → Processing → Analysis → Visualization

Pipeline Purpose

Transform raw data into useful business insights.

Main Pipeline Stages

1. Data Sources
2. Data Ingestion
3. Data Storage
4. Data Processing
5. Query & Analysis
6. Visualization & Reporting



AWS Big Data Pipeline Architecture

Pipeline Stage	AWS Service
Ingestion	Kinesis, Glue
Storage	Amazon S3
Processing	Glue ETL, EMR
Querying	Athena
Visualization	QuickSight

Batch vs Streaming Analytics

Feature	Batch Processing	Streaming Processing
Latency	Minutes to hours	Seconds
Processing Style	Large datasets	Continuous events
Use Cases	Reports, ETL	Fraud detection
Tools	EMR, Hadoop	Kinesis

When to Use Batch Analytics

- Historical reporting
- Data warehousing
- Machine learning training
- Large ETL jobs
- Financial reports

Characteristics

- High accuracy
- Lower cost
- Scheduled processing
- Large-scale computation

When to Use Streaming Analytics

Best For

- Fraud detection
- Live dashboards
- Real-time alerts
- Monitoring systems
- Instant recommendations

Characteristics

- Low latency
- Immediate insights
- Continuous processing

An e-commerce company tracks:

- Product clicks
- Cart actions
- Purchases
- User sessions

Analytics Goals

Real-Time

- Active users now
- Trending products

Historical

- Daily sales reports
- Customer behavior analysis

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AWS Glue, Athena, and Kinesis Family

What is Data Ingestion?

The process of collecting and moving data into analytics systems.

Challenges

- Traffic spikes
- Reliability
- Scalability
- Event ordering

Solution

Use scalable streaming systems like Kinesis.

What is a Data Lake?

A centralized repository storing structured and unstructured data.

AWS Data Lake Foundation: Amazon S3

Types of Data

- Structured
- Semi-structured
- Unstructured

Data Lake Zones: Raw Zone - Original source data

Clean Zone: Validated and transformed data

Curated Zone: Business-ready analytics datasets

Benefits

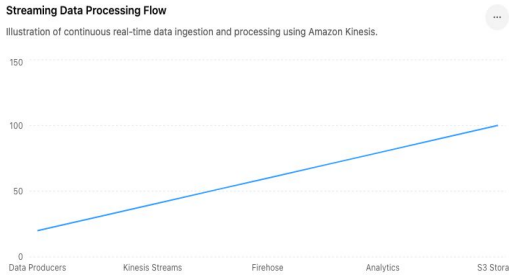
- Durable
- Scalable
- Cost-effective

What is Amazon Kinesis?

Amazon Kinesis is AWS's real-time data streaming service.

Kinesis Services

- Kinesis Data Streams
- Kinesis Firehose
- Kinesis Data Analytics
- Kinesis Video Streams



Key Components

- Producers
- Streams
- Shards
- Consumers

Streaming Pipeline

- Data producers send streaming data
- Kinesis captures streams
- Firehose stores data into S3
- Analytics service processes streams in real time

Advantages

- Fully managed
- Real-time analytics
- Durable storage
- Replay capability

Challenges

- Higher cost at scale
- Shard management
- Increased complexity

Real-Time Analytics Use Cases

- IoT monitoring
- Website clickstream analysis
- Fraud detection
- Social media analytics

The Need for Metadata & Catalogs

Problem: S3 only stores files — not schemas or table definitions.

Metadata Needed

- Columns
- Data types
- Partitions
- Locations

Solution: AWS Glue Data Catalog

What is AWS Glue?

AWS Glue is a fully managed ETL (Extract, Transform, Load) service.

Key Features

- Automatic schema detection
- Data catalog creation
- ETL job automation
- Serverless processing
- Workflow orchestration

Components

- Glue Crawlers
- Glue Data Catalog
- Glue Studio
- Glue Jobs

AWS Glue Architecture

Glue Workflow

1. Upload source data to S3
2. Glue crawler scans data
3. Metadata stored in Data Catalog
4. ETL jobs transform data
5. Output stored in target repository

Benefits

- Reduced manual work
- Faster ETL development
- Better metadata management

Technologies

- Apache Spark
- Python
- Scala

What is AWS Athena?

Athena is a serverless query service that analyzes data directly in Amazon S3 using SQL.

Features

- Serverless architecture
- SQL-based querying
- Works with CSV, JSON, Parquet, ORC
- Cost-effective pricing
- Integration with BI tools

Advantages

- No infrastructure management
- Fast ad hoc analytics
- Easy integration with S3

Athena Query Process

Workflow

1. Data stored in S3
2. Glue Catalog provides metadata
3. Athena runs SQL queries
4. Results stored back in S3

Example Use Cases

- Log analysis
- Data exploration
- Reporting
- Business intelligence

Service	Purpose	Processing Type
Kinesis	Real-time ingestion	Streaming
Glue	ETL & metadata	Batch
Athena	SQL analytics	Interactive

Amazon EMR, Hadoop, and MapReduce

Why Distributed Processing?

Problem: Single machines cannot process terabytes of data efficiently.
Solution: Distributed processing

- Split data
- Process in parallel
- Scale horizontally

Benefits

- Faster computation
- Scalability
- Fault tolerance

Amazon EMR

Amazon EMR is a managed big data processing platform.

Supported Technologies

- Hadoop
- Spark
- Hive
- Presto
- Trino

Key Benefits

- Managed clusters
- Auto scaling
- AWS integration

Core Components

- Master Node
- Core Nodes
- Task Nodes

Optional Components

- Hive
- Spark
- HBase
- Presto
- Flink

EMR Features

- Autoscaling
- Security integration
- Distributed storage

Understanding Hadoop

Hadoop Components

- **HDFS** – Distributed storage system
- **MapReduce** – Distributed processing engine

Hadoop Advantages

- Parallel processing
- Scalability
- Fault tolerance
- High availability

MapReduce Concept

Divide and Conquer Approach

1. **Map Phase**
   - Split data into smaller chunks
   - Process in parallel
2. **Reduce Phase**
   - Aggregate processed results
   - Generate final output

Apache Hive

Hive is a SQL-like query engine for Hadoop.

Features

- Schema-on-read
- Partitioning
- Aggregation
- SQL-like syntax
- User-defined functions

Best Use Cases

- Data warehousing
- Batch analytics
- SQL reporting

Apache Spark

Spark is a fast, in-memory data processing engine.

Key Features

- Fast processing
- Fault tolerance
- Scalability
- Machine learning support
- Streaming analytics

Spark Libraries

- Spark SQL
- Spark Streaming
- MLlib
- GraphX

EMR Big Data Processing Example

Example Workflow

1. Upload CSV dataset to S3
2. Launch EMR cluster
3. Submit Spark job
4. Process 200,000+ records
5. Store results in S3

Outcome

- Faster analytics
- Distributed processing
- Efficient resource usage

Feature	Glue ETL	EMR
Management	Serverless	Cluster management
Complexity	Simple	Advanced
Flexibility	Limited	High
Best For	ETL pipelines	Heavy processing

Visualization with Amazon QuickSight

The Role of Business Intelligence

Why Dashboards Matter: Business users need
- Visual reports
- KPI dashboards
- Interactive analytics
- Mobile access

Goal: Transform data into business insights.

Introduction to QuickSight

Amazon QuickSight is a cloud-based Business Intelligence (BI) service.

Main Features
- Interactive dashboards
- Data visualization
- Data preparation
- Collaboration
- SPICE in-memory engine

SPICE Engine

What is SPICE?
SPICE = Super-fast, Parallel, In-memory Calculation Engine

Benefits
- Faster queries
- Improved dashboard performance
- Reduced latency
- Better scalability

QuickSight Architecture

Workflow
1. Connect datasets from S3/Athena/Redshift
2. Prepare and clean data
3. Create visualizations
4. Build dashboards
5. Share insights

Supported Visualizations
- Bar charts
- Pie charts
- Maps
- Tables
- Line charts

QuickSight Dashboard Example

Example Analytics
- Sales performance
- Customer trends
- Social media engagement
- Financial reporting



Further reading

EIU

- Song, L. (2023). The self-taught cloud computing engineer: A comprehensive professional study guide to AWS, Azure, and GCP (1st ed.). Packt Publishing. Chapter 5 – Amazon Data Analytics Services (Song, 2023, pp. 105–144)
- Amazon Web Services Glue. https://aws.amazon.com/glue/
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- Amazon Web Services EMR. https://aws.amazon.com/emr/
- Amazon Web Services QuickSight. https://aws.amazon.com/quick/quicksight/

Thank you!

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